Amirhossein Kazerouni

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Research Summary

PhD researcher working on generative models and 3D/4D reconstruction. Current work includes a geometry foundation model for panoramic video with feed-forward Gaussian reconstruction (Samsung AI Center Toronto) and 4D face generation controllable through facial action units. First- and co-first-author papers at NeurIPS, CVPR, ICCV, and WACV; Samsung Best Paper Award (Bronze) for Face2Scene.

Education

University of Toronto — Toronto, Canada
Ph.D. in Computer Science, advised by Babak Taati and Michael Brudno — *Jan. 2024 – Present*

Iran University of Science and Technology (IUST) — Tehran, Iran
B.Sc. in Electrical Engineering (Control) — *Sep. 2017 – Feb. 2022*

Research Experience

Samsung AI Center Toronto — Toronto, Canada
Research Intern. Supervisors: Alex Levinshtein, Iqbal Mohomed, Konstantinos G. Derpanis — *May 2025 – Present*

- **Panoramic 4D Reconstruction:** Developing a geometry foundation model for panoramic video, and a compact feed-forward Gaussian reconstruction model built on its estimated geometry.
- **Face2Scene:** Estimates the degradation from a low-/high-quality face pair, with the high-quality face from reference-based face restoration, to guide one-step diffusion restoration of the whole scene. *CVPR 2026.*
- **AVIS:** Developed a deployment-friendly policy that prunes redundant visual tokens and spends the saved compute on more self-consistency rollouts, improving VLM accuracy at matched inference cost. *NeurIPS 2026.*
- Contributed to BurstGP, raw burst image super-resolution with generative priors. *arXiv 2026.*

University of Toronto & Vector Institute — Toronto, Canada
Graduate Research Assistant. Supervisors: Babak Taati, Michael Brudno — *Jan. 2024 – Present*

- **Controllable 4D Generation:** Using a video diffusion model to estimate a person's 3D face in a neutral expression from a single image, then controlling its expressions with facial action units (AUs).
- **LIFT:** Developed a meta-learned encoder for images and 3D shapes using parallel local implicit functions and hierarchical latents, used for generative modeling and classification at lower compute. *ICCV 2025.*
- **SUM:** Developed a single Mamba-based saliency model that adapts to different image types instead of one model per domain, to better model human visual attention. *WACV 2025, oral.*

RWTH Aachen University — Aachen, Germany
Research Assistant. Supervisors: Dorit Merhof, Reza Azad — *2022 – 2024*

- **Medical Image Segmentation:** Built 2D/3D CNN–Transformer and diffusion models with multi-scale fusion, frequency attention, self-supervision, and boundary-aware training. *WACV, MICCAI, ISBI, MIDL (2 orals).*
- **INCODE:** Developed a harmonizer network that uses deep prior knowledge to tune the sinusoidal activations of an implicit neural representation, improving signal fitting, denoising, and super-resolution. *WACV 2024.*

DGSculptor — Montreal, Canada
Machine Learning and Computer Vision Researcher. Supervisor: Alireza Morsali — *2021 – 2023*

- **Ensemble INRs:** Co-developed an ensemble implicit neural representation in which independent tiny MLPs each fit part of the signal, reaching higher PSNR than SIREN with fewer FLOPs.

IUST AI and Control Lab — Tehran, Iran
Computer Vision Researcher (undergraduate) — *2019 – 2021*

- Developed a fusion-based multi-stage method for 3D shape analysis of vehicles with a stereo camera, used to measure load protrusion.

University of Tehran Tehran, Iran
Computer Vision Researcher. Supervisor: Mohammad Ali Akhaee *2019 – 2020*

- **Football Analysis:** Built a bird's-eye-view module: a GAN extracts field lines from broadcast frames, and the homography estimated from them maps each frame onto a top-down pitch template. [Project] [Code]

Selected Publications

* equal contribution. Full list on Google Scholar.

- **Face2Scene: Using Facial Degradation as an Oracle for Diffusion-Based Scene Restoration**
  Amirhossein Kazerouni, Maitreya Suin, Tristan Ty Aumentado-Armstrong, Sina Honari, Amanpreet Walia, Iqbal Mohomed, Konstantinos G. Derpanis, Babak Taati, Alex Levinshtein. *CVPR 2026* **SBPA Bronze Award** [Project]
- **AVIS: Adaptive Test-Time Scaling for Vision-Language Models**
  Ahmadreza Jeddi*, Minh Ngoc Le*, **Amirhossein Kazerouni***, Hakki Can Karaimer, Hue Nguyen, Iqbal Mohomed, Michael Brudno, Alex Levinshtein, Konstantinos G. Derpanis, Babak Taati, Radek Grzeszczuk. *NeurIPS 2026* [Project]
- **LIFT: Latent Implicit Functions for Task- and Data-Agnostic Encoding**
  Amirhossein Kazerouni, Soroush Mehraban, Michael Brudno, Babak Taati. *ICCV 2025* [Project]
- **SUM: Saliency Unification through Mamba for Visual Attention Modeling**
  Alireza Hosseini*, **Amirhossein Kazerouni***, Saeed Akhavan, Michael Brudno, Babak Taati. *WACV 2025* **Oral** [Project]
- **INCODE: Implicit Neural Conditioning with Prior Knowledge Embeddings**
  Amirhossein Kazerouni, Reza Azad, Alireza Hosseini, Dorit Merhof, Ulas Bagci. *WACV 2024* [Project]
- **PriLoRA: Prior-Conditioned Low-Rank Adapters for Medical Vision Models**
  Amirhossein Kazerouni*, Sanaz Karimijafarbigloo*, Ali Aghayari*, Ali Ghasemzadeh Khorzoughi*, Reza Azad, Wojciech Samek, Dorit Merhof, Michael Brudno, Babak Taati. *MICCAI 2026* [Code]
- **HiFormer: Hierarchical Multi-Scale Representations Using Transformers for Medical Image Segmentation**
  Moein Heidari*, **Amirhossein Kazerouni***, Milad Soltany*, et al. *WACV 2023* [Code]
- **OpenVAM: Open-World Visual Attention Modeling with VLMs**
  Kiana Hooshanfar*, **Amirhossein Kazerouni***, Alireza Hosseini*, Michael Brudno, Babak Taati. *WACV 2027* [Project]
- **A Unified Theory of Sinusoidal Activation Families for Implicit Neural Representations**
  Alireza Morsali, MohammadJavad Vaez, Mohammad Hossein Soltani, **Amirhossein Kazerouni**, et al. *Transactions on Machine Learning Research (TMLR), 2026* [Project]

Honors and Awards

2026	**SBPA Bronze Award** (Samsung Best Paper Awards, Samsung Group-wide), for Face2Scene.
2026	**Outstanding Reviewer**, ECCV 2026.
2024–	**Vector Research Grant**, Vector Institute.
2021	**1st place (team)**, national Rahneshan autonomous vehicle competition, INEF, Iran.
2021	**2nd place (team)**, FIRA World Cup, Autonomous Cars League.

Academic Service

Conference reviewer	CVPR 2026, ECCV 2026 (Outstanding Reviewer), NeurIPS 2024–2026, AAAI 2027, MICCAI 2024, 2026
Journal reviewer	Medical Image Analysis; Engineering Applications of Artificial Intelligence

Teaching and Mentoring

Teaching Assistant	University of Toronto: CSC2503 Computational Vision; CSC420 Image Understanding (3 terms); CSC384 Artificial Intelligence; CSC148 Computer Science
Co-founder	AIR (Artificial Intelligence and Robotics) Center, Tehran (2020–2022), a training center offering courses in deep learning, machine learning, and Python

Skills

Programming	Python, C/C++, MATLAB; PyTorch, JAX, TensorFlow, OpenCV, NumPy; Linux, Git